

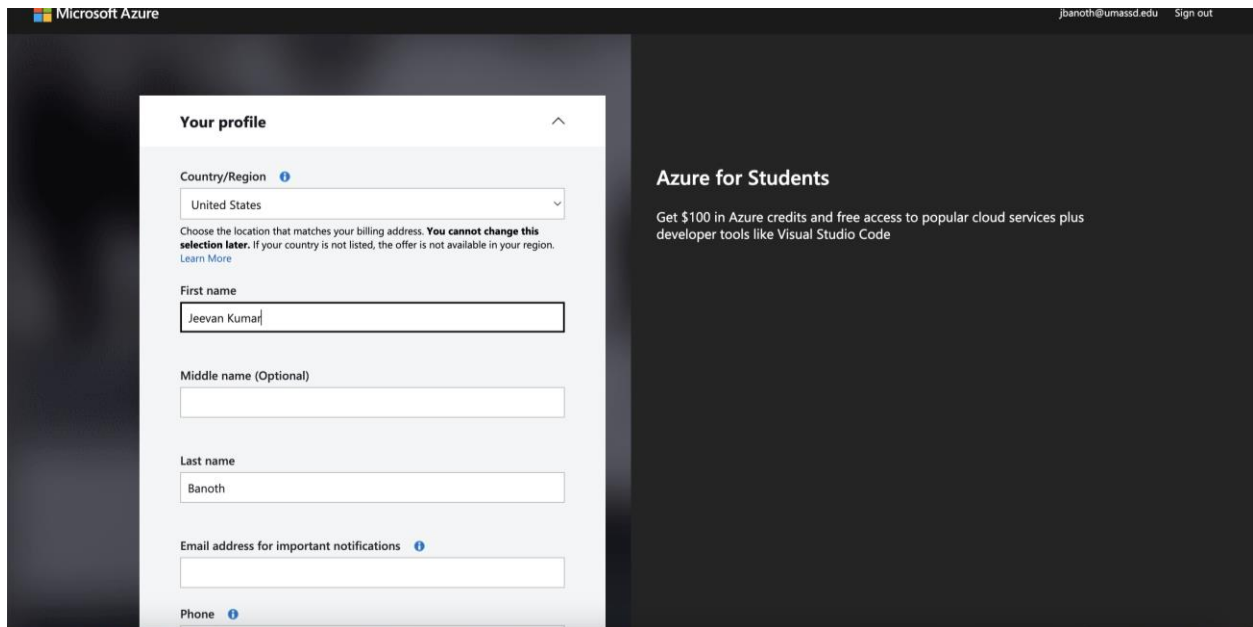
Jeevan Kumar Banoth – 02105145

CIS 552: Database Design

Homework – 4:

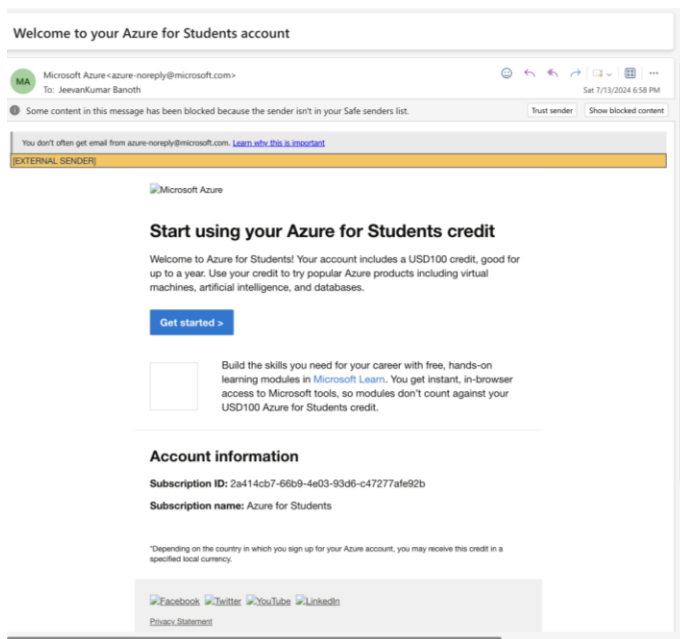
Task 1:

When working on a Microsoft Azure system, you should include these common details: your first name; last name; mobile number and email address.



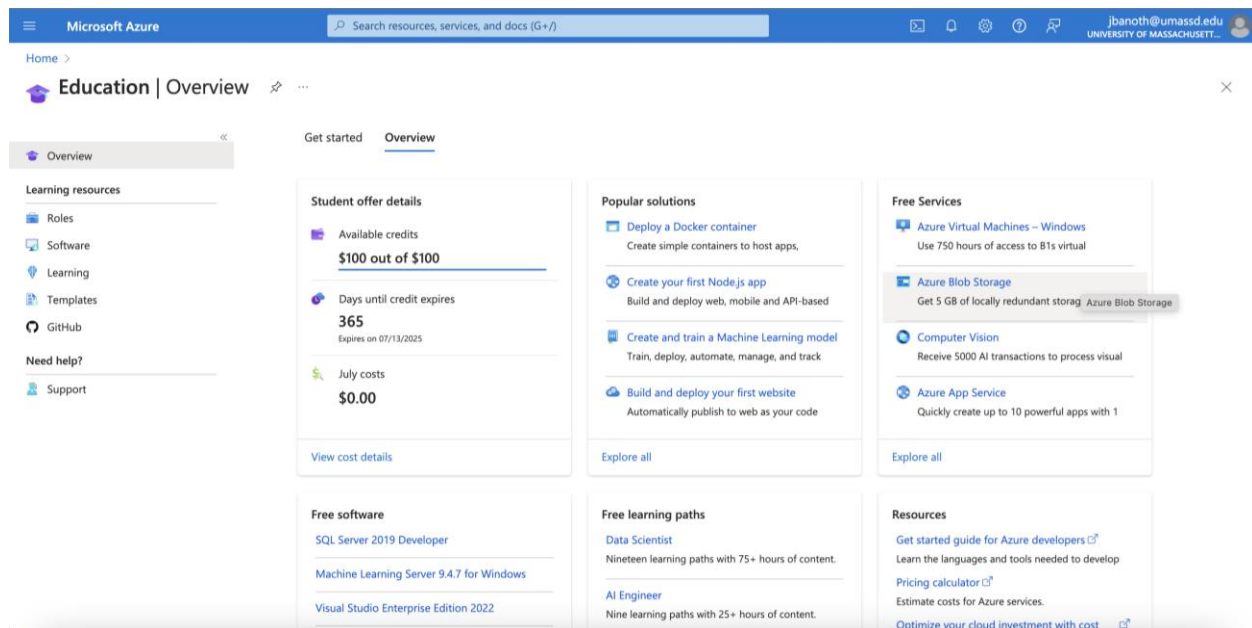
The screenshot shows the 'Your profile' form in the Microsoft Azure portal. The form is set against a dark background. On the right side, there is a section titled 'Azure for Students' with the text: 'Get \$100 in Azure credits and free access to popular cloud services plus developer tools like Visual Studio Code'. The form fields include: 'Country/Region' (a dropdown menu currently showing 'United States'), 'First name' (text input with 'Jeevan Kumar'), 'Middle name (Optional)' (empty text input), 'Last name' (text input with 'Banoth'), 'Email address for important notifications' (empty text input), and 'Phone' (empty text input). A 'Sign out' link is visible in the top right corner of the page.

We will receive an email about the Microsoft Azure membership when I submit my details, at which point we can move on to the next phase.

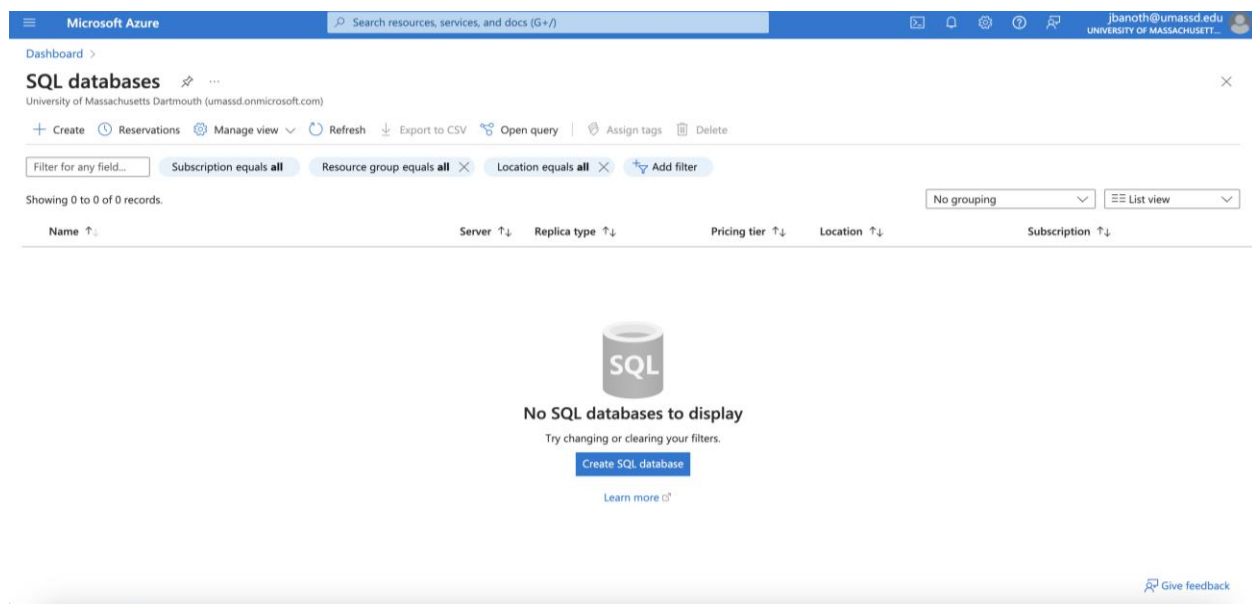


Task 2: Establish and Set Up a Fresh SQL Server Database

We will receive a dashboard, or home, as shown below, after establishing an account for free Azure cloud services. Select "portal" if not. In the cloud, we are now creating a new database server, a resource group, and an administrative username and password for the database server.



The main page of SQL databases will appear when you search for "SQL database" in the search tab, as seen below.



The database construction procedure can then be initiated by choosing the Generate SQL Database button. We must now supply the necessary information. Since I'm using my account, I created a resource group and gave it the name jeevanrishi.

Microsoft Azure

Search resources, services, and docs (G+)

Dashboard > SQL databases > Create SQL Database >

Create SQL Database Server

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name * .database.windows.net

Location *

Authentication

Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) or using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

Authentication method ☒ Use Microsoft Entra-only authentication

OK

Providing the server and database names is the next step. We should switch to SQL server authentication for authentication. and create a password and username. It takes time to deploy after creation.

The database server is now ready to operate on the cloud. Subsequently, the SQL database server is ready to be used in Azure Cloud.

Microsoft Azure

Search resources, services, and docs (G+)

Dashboard >

Microsoft.SQLDatabase.newDatabaseNewServer_3cd8007c1e99469da3af5 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServe... Start time : 13/7/2024, 8:02:53 pm

Subscription : Azure for Students Correlation ID : 0749ab83-d173-4774-86a7-f09a14015b15

Resource group : umass_student

Deployment details

Resource	Type	Status	Operation details
jeevanrishi0/jeevan	Microsoft.Sql/servers/databases	Accepted	Operation details
jeevanrishi0/Default	Microsoft.Sql/servers/sqlVulnera	Created	Operation details
jeevanrishi0/Default	Microsoft.Sql/servers/connectio	OK	Operation details
jeevanrishi0	Microsoft.Sql/servers	OK	Operation details
jeevanrishi0	Microsoft.Sql/servers	Created	Operation details

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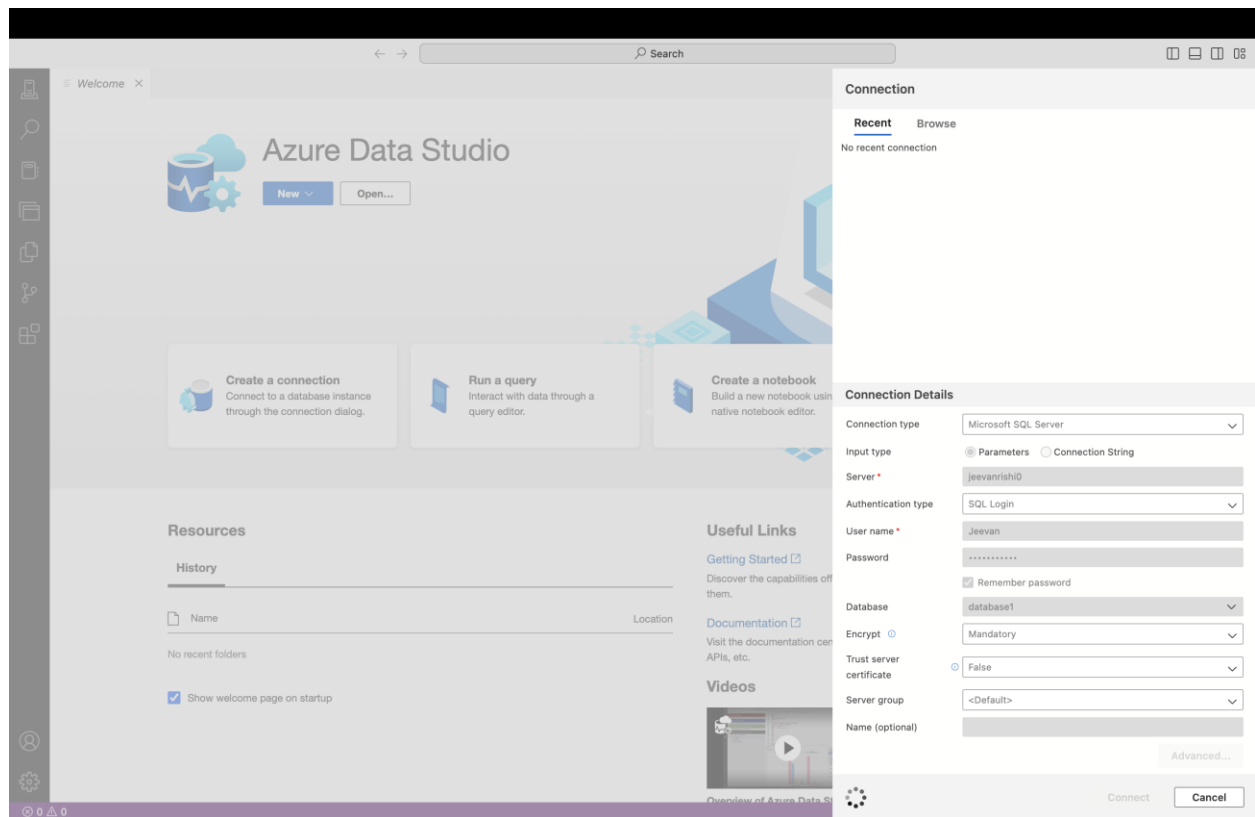
Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

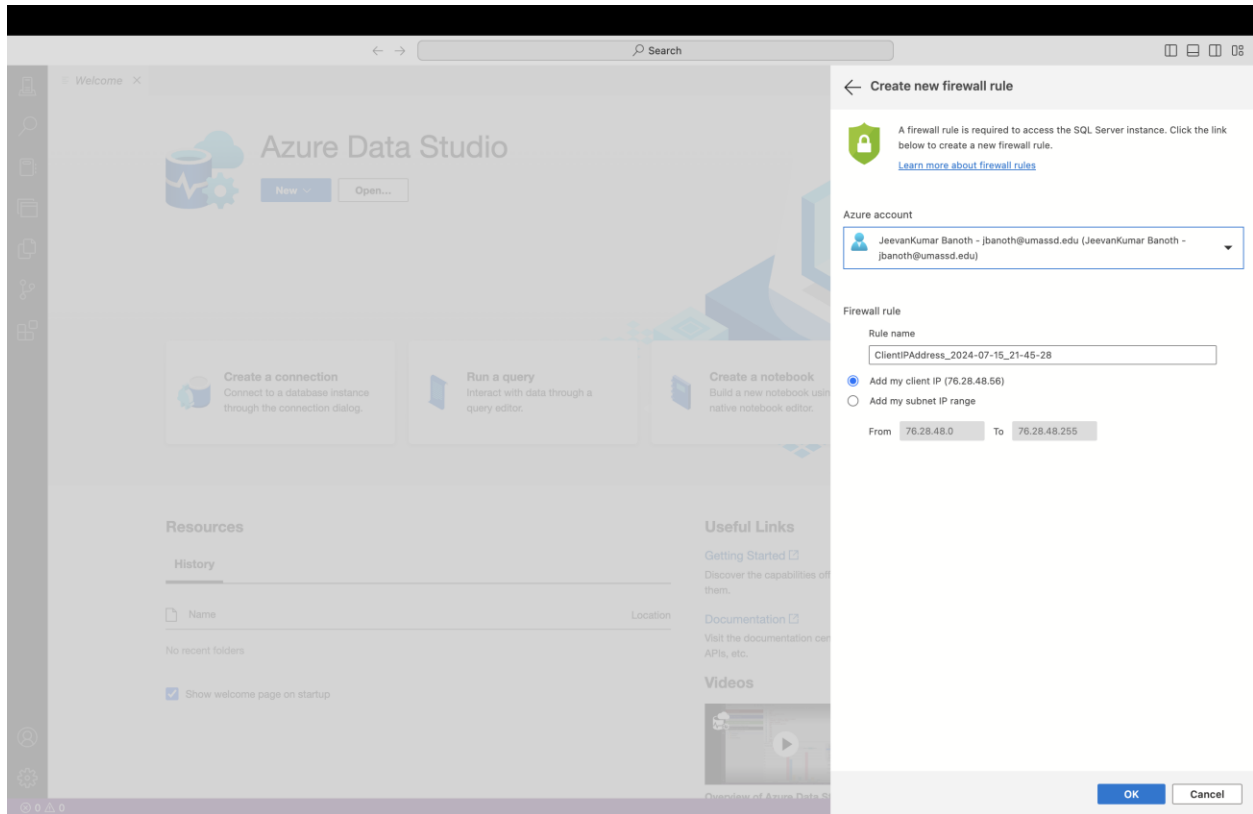
[Find an Azure expert >](#)

Task 3: Utilize Azure Data Studio to create tables, add information, and run queries.

We need to open Microsoft Server Management Studio to connect to the database.



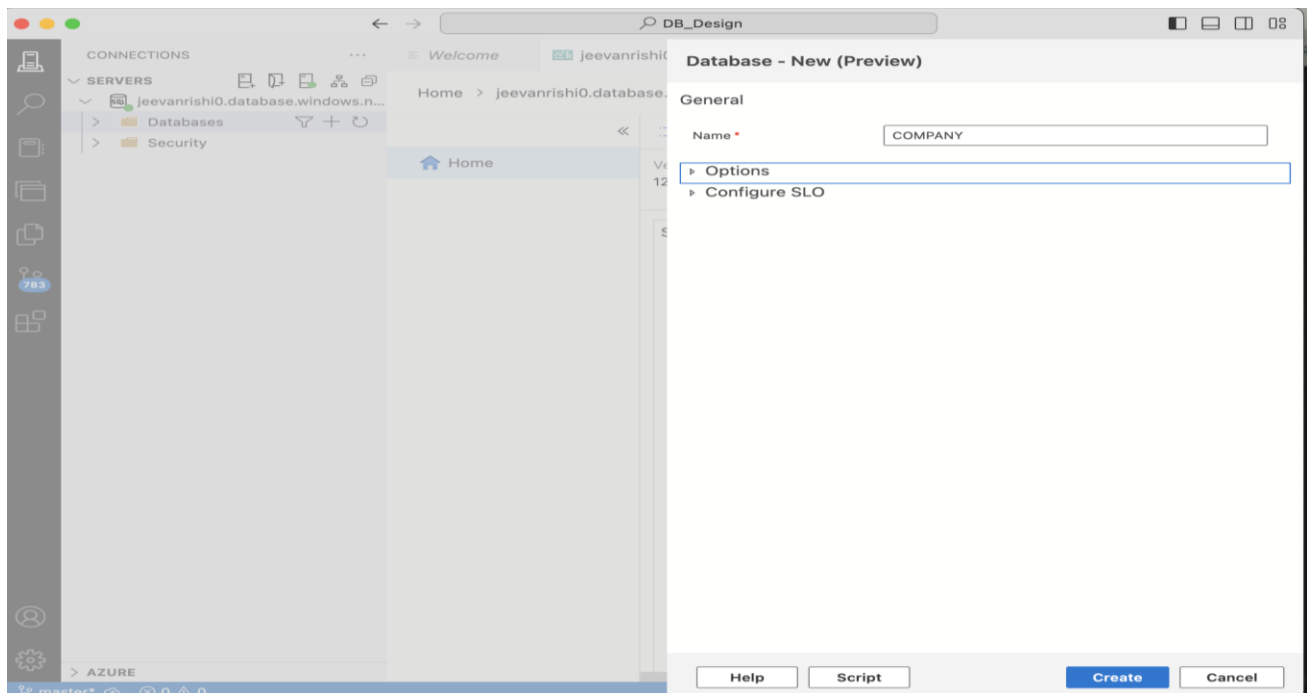
To connect to the database server, utilize the jeevanrishi connect button. You can select Database Engine from the dropdown, then establish the connection to the Azure database server as follows. The address of my server is jeevanrishi.database.windows.net



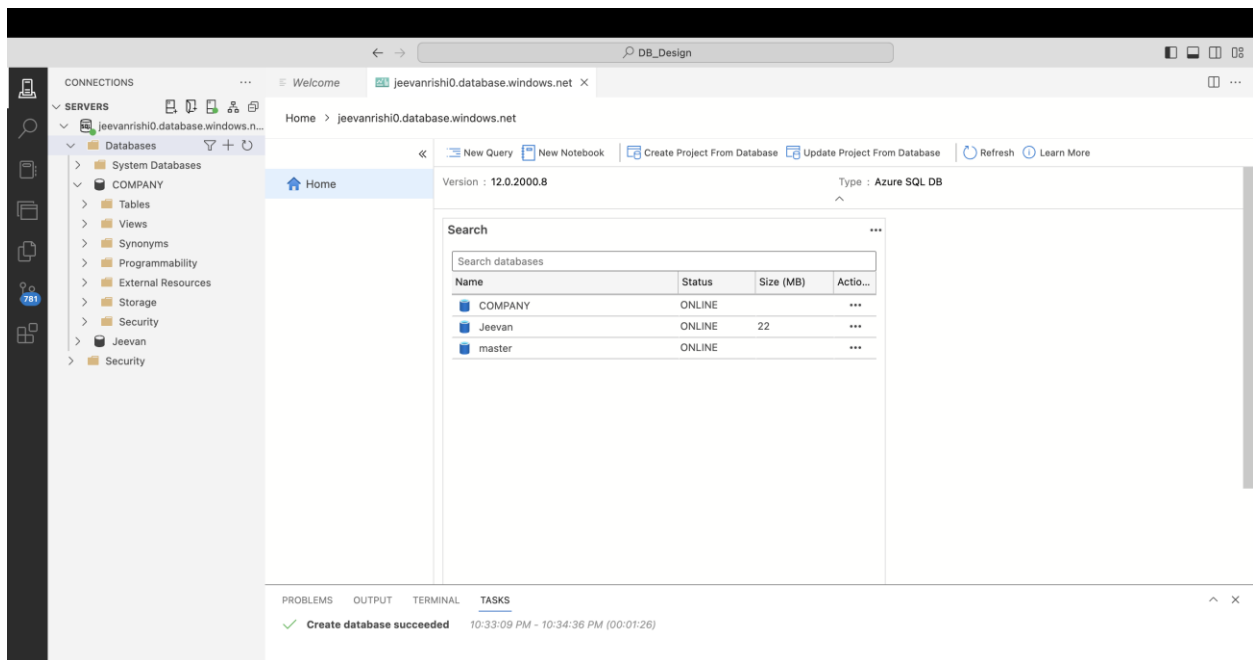
The username and password must be used as login credentials to connect the database server to SQL authentication.

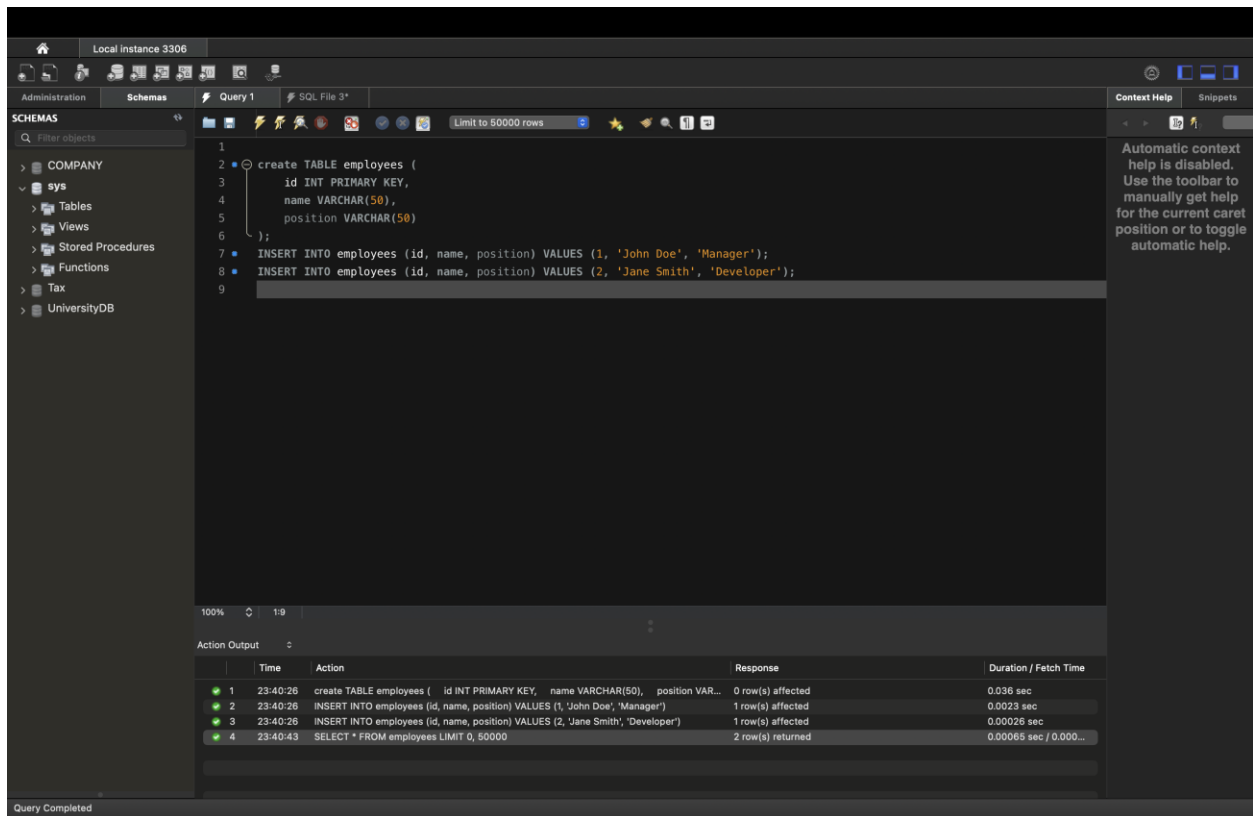
On the cloud jeevan, we can now create tables, add data, and execute queries¹. In addition, we may deploy your COMPANY database using a local laptop database server hosted in the cloud.

Task 4: Install or move the "COMPANY" database from the local database server to the SQL server in the Azure cloud:

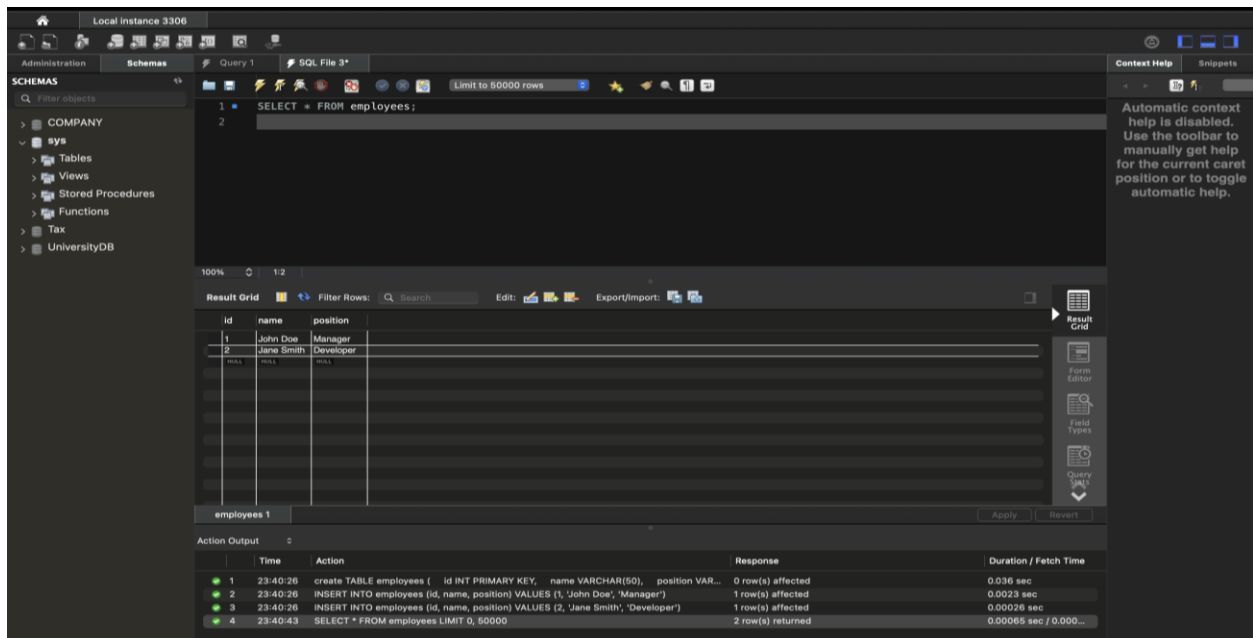


As you can see, here we have created a database named COMPANY using Azure Data studio.

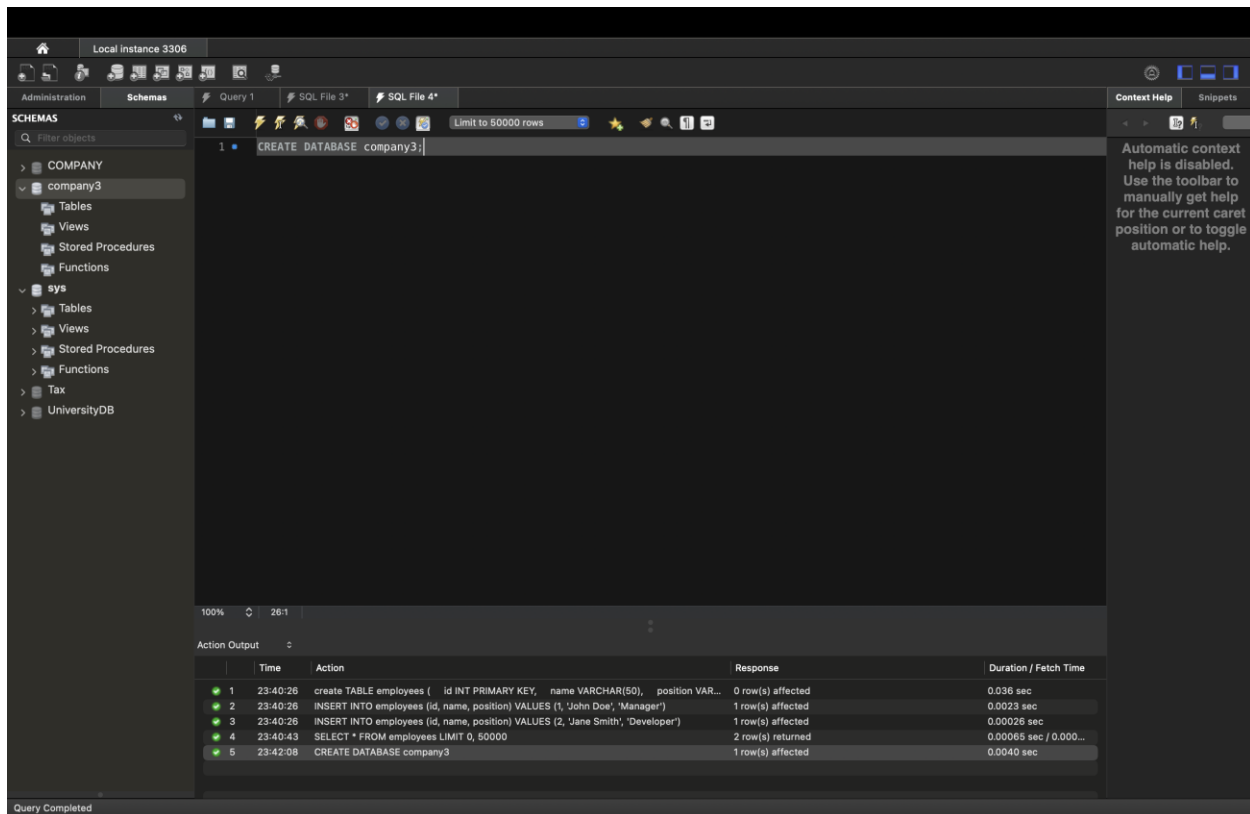




Using the SQL workbench, we have executed the code above and created tables.



As you can see this is the outcome of the code and tables as shown above.



For creating the database, we can also do it by using the above SQL code and you can see that company 3 has been created.

After refreshing, we can see the new database COMPANY-3.

Conclusion:

I started my adventure with Azure when I created an account and entered my personal data, saying that I would buy, which then paved the way for some more tasks around databases. Having entered Azure portal, I made a new SQL Database Server on a particular resource group and specified necessary security settings. SQL Server authentication was my choice because it offers secured methods of interacting with a database later.

Once I installed Azure Data studio, I connected to the Azure-hosted database server. Introducing our “COMPANY” database from our on-premises setup into the Azure cloud is what I would term a hands-on experience that showed me how flexible and elastic cloud service provision can be. Right after its movement happened, I took an active role in managing this database and coming up with its tables through certain queries that ran against it.

The creation of the "COMPANY-3" database was particularly noteworthy since it demonstrated the practical application of SQL scripts and the speed at which Azure could process database queries.

The insights came from the tasks taken were many-this demonstrated that cloud held a central place in today's data handling methods beyond what Azure database services could handle as just part. Before coming out of the lab, I had a database operating on the cloud which showed how effective both their tools and procedures were, for instance that of Microsoft Azure. The findings from querying "Company" databases verified that this system could reproduce what was found in there.